Mill Parameter Archiving User's Guide

Order Number: GEK-97001

July 1, 1989

This document details the use of the Mill Parameter Archiving (MPA) function for data collection and analysis. Any GE Control System Freeway (CSF) global variable can be sampled; up to 250 signals may be sampled once every second, and an additional 250 signals may be sampled every minute. The sampled data is saved on the supervisory computer disk. A function is provided which provides correlation statistics and graphical display of the sampled data.

Richard E. Gillmore & Michael R. Good

GE Drive Systems
1501 Roanoke Boulevard
Salem, Virginia 24153

Contents

1 DESCRIPTION 1

2 REQUIREMENTS 1

3 OPERATOR INTERFACE 2
- 3.1 PLOT Specification Language 2
  - 3.1.1 Sample Plot Descriptor File • 2
  - 3.1.2 Plot File Commands • 3
- 3.2 Graphical Analysis 5
  - 3.2.1 Purpose • 5
  - 3.2.1.1 Available Commands • 6
  - 3.2.1.1.1 MAIN SCREEN • 6
  - 3.2.1.1.2 EXPANDED MODE • 7
  - 3.2.1.1.3 DISPLAY NUMBERS MODE • 8
  - 3.2.1.1.4 STATISTICS MODE • 8
- 3.3 Data Collection 10

4 MAINTENANCE 11
- 4.1 Data Collection Function 11
- 4.2 Data Analysis Function 12

5 CONFIGURATION 12

APPENDIX A INSTALLATION A-1

1 Description

The Mill Parameter Archiving (MPA) function provides data collection for comparison and analysis. Any CSF global variable can be sampled; up to 250 signals may be sampled once every second, and an additional 250 signals may be sampled every minute. The sampled data is saved on the supervisory computer disk. A function is provided which provides correlation statistics and graphical display of the sampled data.

MPA is a diagnostic tool intended to provide historical information about mill parameters over a fairly long period of time (e.g. a week). This data can then be used to help locate areas of degrading performance and predict possible equipment failures.

The MPA system has two major parts. An on line function resident in the supervisory VAX is responsible for collecting data from the CSF and creating the data files. A second interactive function then processes these data files and provides graphical displays and correlation statistics.

MPA is not required for operation of the mill. The data collection and analysis in the supervisory computer is completely independent of the control system.

2 Requirements

Graphics Terminal

The graphical analysis tool requires a color graphics terminal. Currently supported devices are DEC's VT340, and the IDT 2250.

Control System Freeway

The MPA software collects data variables resident on the Control System Freeway (CSF). The associated CSF hardware and software must be present for MPA to function.

Data Storage

MPA allows for up to 250 variables to be sampled every second, and another 250 variables sampled every minute. Approximately 7 days of data may be collected. Assuming each sampled variable is 4 bytes long, this function requires 615 Mbytes of disk storage. A formatted RA82 disk holds 623 Mbytes of data, so this function requires a dedicated RA82 disk to store the data files.

$$\left(250 \frac{\text{smpl}}{\text{min}} + \left(250 \frac{\text{smpl}}{\text{sec}} * 60 \frac{\text{sec}}{\text{min}} \right) \right) * 1440 \frac{\text{min}}{\text{day}} = 21960000 \frac{\text{samples}}{\text{day}}$$

$$21960000 \frac{\text{smpl}}{\text{day}} * 7 \text{ days} * 4 \frac{\text{bytes}}{\text{smpl}} = 615 \text{ Mbytes}$$

Note that this does not account for any additional storage for file header information and other RMS overhead.

3 Operator Interface

The MPA system has two major parts. The data collector is responsible for collecting IOCOM data from the CSF. The second function then converts the collected data into graphical plots and displays them. It is this second function which provides the primary operator interface to the "mill parameter" data. The following sections detail how the operator can specify a plot, and how to use the provided PLOT utility to graphically analyze the data.

3.1 PLOT Specification Language

This section details how to create a plot specification file, which is used to specify plotting screens for the PLOT utility.

The user can define which variables are displayed together on each screen by creating input data files for the PLOT analysis function. Each screen may contain a practical limit of four plots, and each plot may contain up to four variables each (or up to 8 logicals), for a total of 16 real data variables (or 32 logical variables) on a single screen.

The user can interactively define the starting and ending date and time for the plot, within the range of data points which are present on the disk.

3.1.1 Sample Plot Descriptor File

The following is a sample of a plot descriptor file. The file is free-format; each line may contain any number of tabs and/or spaces between keywords and parameters and comments. Blank lines are permitted for readability, and any text following a "!" character is ignored (for comments). Some of the commands are optional, as noted in the file comments.

All lowercase text in the example denotes parameters which the user must define.

```
TITLE "plot_title"              ! optional (defaults to "plot_name")
X_SCALE min max "units"         ! optional (program-supplied defaults)
SAMPLES count                   ! optional (program-supplied defaults)

PLOT_1 min max units            ! required (units optional)
  DATA name                     ! required
  DATA name "description"       ! required (description optional)
!       .
!       .   (up to four per real plot)
!       .   (if min & max are both 0.0, then program will autoscale)
```

```
PLOT_2 Logical                  ! required
  DATA name                     ! required
  DATA name                     ! required
!      .
!      .   (up to eight per logical plot)
DISPLAY PLOT                    ! required to actually draw the plots
INCLUDE plot_name               ! to chain to another file
! ------------------------------------------------------------------------
```

3.1.2 Plot File Commands

The following commands are supported in a plot definition file:

- TITLE "text"

  This command defines a title for the graph. If you do not specify a title, the program uses the name of the plot definition file as the title. If your title contains lower case text, you must enclose it in quotes or it will be converted to upper case.

- X_SCALE min max units NOW

  This command defines the scaling for the x-axis. "min" and "max" are real numbers which define the range of times you wish to sample. The "units" may be "Days", "Hours", "Minutes", or "Seconds", which may be abbreviated to a single character if desired. The times specified are all relative to midnight of the previous night, unless the optional keyword NOW is specified, in which case they are relative to the current time. For example, to view data for yesterday, you could specify the command "XSCALE -1.0 0.0 Days".

- SAMPLE n

  This command is used to determine the number of samples to read from the file. If you specify "n" as 0, all samples will be read and plotted. If you are viewing data for an entire week, this means over 600,000 points must be plotted, which will take a considerable amount of time. If "n" were specified as 1000, every 600th point would be used for this example. Note that the program rounds "n" up to the next integral multiple of the number of actual points.

- PLOT_n min max "units"

  This command starts a real subplot (one box on the screen). Subplots must be defined in order (1-4). The parameters define the y-axis scaling for the subplot (real values). Specifying "min" and "max" as both 0.0 will cause the plotting routine to select appropriate values so that all the data is displayed (autoscaling). Up to four real data points can be specified in each subplot. The "units" qualifier specifies text for the y-axis of the plot.

- PLOT_n LOGICAL

  This command starts a logical subplot. Subplots must be defined in order (1-4), with any combination of real and logical subplots together on a screen. There are no scaling values for a logical subplot; only the keyword "LOGICAL", which may be abbreviated to a single "L" if desired. Up to eight logical data points can be specified in each subplot.

- DATA variable_name "description"

  The DATA statement defines a variable to be plotted. This selects a variable as well as specifying an optional description with which to tag each plot curve. The format of the variable name is "F.name" or "S.name", where "F" means a fast (once-per-second) point and "S" means a slow (once-per-minute) point. The "name" portion must exactly match the CSF variable name of a data point being collected. If the "description" field is omitted, the variable name itself is printed to the right of each curve.

- DISPLAY PLOT

  This command causes the plot to be drawn on the screen. This must be included in at least one plot file to create a plot. It allows several screens to be specified in one plot file, or several plot files to be chained together.

- INCLUDE plot_file

  This command causes the parser to begin reading from the defined plot file. The ".PLOT" file extension is assumed if no extension is present. This command allows you to "nest" plot definitions as desired. For example, if you have several single plots defined, you could define a file which contained a list of "INCLUDE" statements which would then cause the plots to be drawn one after the other. The user must select the "P" option in each plot to advance to the next plot in the list.

- DEBUG { ON / OFF }

  This command should normally not be included in a plot descriptor file. It provides additional information about every line of input that is read from the plot descriptor file, and about all the data that is read from the data collection files.

The user can also control the defaults which are applied to any parameters which are not specified in a plot definition file. There is a file called "_DEFAULTS.PLOT" in the MPA_PLOT$Dir: directory; it follows the same format as a plot file, and is read by the parser to initialize each plot.

3.2 Graphical Analysis

The MPA Data Analysis function is provided to display and analyze the data which has been collected. The data is displayed graphically on an IDT or VT340 terminal; the user can define which variables are displayed together on each screen by creating input data files for the analysis function. Each screen may contain up to four plots, and each plot may contain up to four variables each (or up to 8 logical), for a total of 16 real data points (or 32 logical points) on a single screen.

The user can interactively define the starting and ending date and time for the plot, within the range of data points which are still present on the disk. Data for the current day may also be plotted, with data available through the current time.

Once a time interval has been selected, the function determines the optimal number of data samples to take from the database so that the user gets as much information as possible and the plot can be generated in a timely manner. The user also has the option to specify the number of samples in the plot definition file, if desired.

The plotting function is an interactive utility which uses menus and prompts to lead the user through the process of generating a set of plots of the various data values which have been collected. The plots are labeled with the names of each variable plotted and the date and time along the x-axis.

The user is presented with a menu of predefined plots (limit of 64). Plots can be defined by creating a plot definition file (using any DEC editor), which defines the variables to be plotted together on a single screen, some scaling information, and other plot parameters. The name of the plot definition file is the "name" of that plot screen; the user may create up to 64 of these plot files. The capability to "chain" several plot files together to form a logical sequence of screens is also provided.

Since this function is an interactive utility, all errors are displayed to the user at runtime. If a pre-defined plot file requests plotting of some data point which was not collected, the corresponding plot on the display will be blank and contain an appropriate error message.

The following sections describe the actual PLOT utility with which the user interactively analyses their plotting data. The purpose and function of each interactive command is presented.

3.2.1 Purpose

The PLOT utility is designed to provide a data analysis and plotting function in one easy to use utility. User interaction is crucial to allowing flexible data analysis. A statistics mode of operation is provided wherein the user is presented with a list of all non-logical variables from the main plotting screen. The user selects two variables to generate cross-correlation statistics, or the same variable is selected twice to perform auto-correlation statistics. Two subplots are used to graphically present the two selected variables to the user. A dotted red line is displayed on the same subplot with each curve to depict the regression line resulting from the scaled data. For nonlinear datasets, the regression line (and its coefficients) change as the user changes the plot x-axis scaling. This allows for choosing the exact

time interval in which to analyze the data. On either the main screen, the expanded mode, or the statistics mode, the user may select a function where the data points (x and y values) are tabulated to the screen. This helps to prevent misinterpretation of the data.

3.2.1.1 Available Commands

The available commands are presented according to what "mode" the PLOT utility is in. Two features have been added to aid in the entering of commands. Many prompts display a default value along side the prompt. If the user wishes to accept the default value, merely press the RETURN key. If a new value is to be entered, enter the value before pressing RETURN and the default value is overridden. The second feature added to the interactive plot analysis function is the ability to perform "type-ahead". If a long, slow plot is being displayed, for example, and the user wishes to interrupt this action with a new command, simply press the indicated key for that command (e.g. "D" will enter display numbers mode) and after a brief delay the existing command will be replaced by the new one without waiting for the original command to complete. Users will appreciate the time-saving aspects of this feature. Commands associated with the main screen are described first, followed by commands accessible from the expanded mode, the display numbers mode, and finally the statistics mode.

3.2.1.1.1 MAIN SCREEN

The main screen is the screen the user first sees when entering the PLOT utility invoking a particular plotting file (see the section on the PLOT specification language). This screen limits the number of variables the user may simultaneously analyze. The following commands are accessible from this main screen.

Help Option

This option clears the current plotting screen and displays a list of the commands available from the main screen, along with a short description of each. After reviewing the text, the user may press any key and the screen is returned and re-drawn in the mode from which the HELP option was requested.

Change Scaling

This option allows changing the X or Y scaling, or resetting the plot to the original scale defined in the plot specification file. On the main screen, all subplots have the same x-axis scaling, to allow comparing curves at identical time intervals. Therefore, if the user changes the x-axis by specifying a new XMIN and XMAX (units of time), all subplots are re-displayed when the user presses the UPDATE key. If, however, the user wishes to change y-axis scaling, the PLOT utility prompts the user for which subplot to change, since subplots with differing y-axis units may be displayed on the same plotting screen. Scaling changes do not immediately refresh the screen. See the following description of the "Update Display" function.

Update Display

Scaling changes do not immediately result in an updated plot. This is done to allow users to make multiple scaling changes before waiting for the sometimes tedious graphics screen refresh process to complete. After the user has made the desired scale changes, they simply press the "U" key to update the display using the scaling as changed.

Expanded Mode

Pressing the "X" key is used to enter the expanded mode. In expanded mode, the user is prompted for one subplot (subplots are numbered 1-4 from the top of the screen). After entering their choice, the screen is cleared and the selected subplot is drawn to fill the screen. This option is useful only if multiple subplots are on the original plotting screen, and the user needs more graphics resolution to analyze a plot. The expanded mode plot is drawn with more grid lines (and labeled x and y-axis tick marks) than in the normal, smaller subplots.

To exit from the expanded mode, the user presses "E" for Exit.

Display Numbers Mode

Entering a "D" from the main menu places the user in display numbers mode, where (x,y) data values are displayed in a tabular format. See section Section 3.2.1.1.3 for a description of this mode.

Statistics Mode

Entering an "S" from the main menu places the user in the statistics mode. This mode, and the associated commands, are displayed in section Section 3.2.1.1.4.

Next Plot File

Entering a "P" exits the user from the control of the current plotting "screen" and, if multiple screens are defined for the current plot specification file, proceeds immediately to the next specified plotting screen. If no further screen is specified, this key returns the user to the prompt for a new plot file name.

Abort to Menu

Pressing the "M" key immediately aborts the PLOT utility session, leaving the MPA utility.

3.2.1.1.2 EXPANDED MODE

Pressing the "X" key from the main screen places the user in the expanded mode, as described above. The available commands in this mode are the same as in the main menu, with the following exception.

Exit

"E" exits the expanded subplot mode and returns to the main screen.

3.2.1.1.3 DISPLAY NUMBERS MODE

At any time, in the main screen or expanded mode, the user may enter a mode in which the x and y data values are drawn to the screen in tabular format. The display numbers mode first prompts the user for which subplot to display numbers. After the user responds, the graphics screen is cleared and a small representation of the chosen subplot is drawn to the top of the new graphics screen. Then, using color coding based on the color with which each variable's curve is drawn, the x & y data pairs are displayed on the screen. For each variable, the screen displays 100 (x,y) data pairs at a time and pauses until the user presses any key. For subplots with multiple variables, the next variable is displayed in its color after the numbers from the first variable have been displayed.

The user has control over how the numbers are displayed. To immediately scroll to the next "block" of 100 numbers, the user may press the enter key before the data has been drawn. If the user wishes to immediately proceed to the next variable on a subplot, the "N" key may be pressed. If the user is ready to exit the display numbers mode, the "E" key may be pressed.

3.2.1.1.4 STATISTICS MODE

The statistics mode has been designed separately from the main screen mode to try to provide similar functionality, while at the same time maximizing the ease with which a user can perform auto and cross-correlations on data.

The two curves being analyzed are displayed in two subplots at the top of the plotting screen. A linear regression line is fitted to the data of each curve, and displayed as a dotted red line "underneath" the curve. The coefficients for this line (slope and y-intercept) are displayed to the right of each subplot. Should the number of points plotted fall below 50 points, a small box will be drawn at the location of the SAMPLED data points for that curve, and a cubic spline curve fitting algorithm is then used to generate a solid red curve which passes through each of the SAMPLED data points. The term "SAMPLED" refers to the algorithm used to obtain a one-to-one correspondence between data points from two curves. If one curve samples data every 30 milliseconds for example, and the second curve samples data every 50 milliseconds, then the data points will not be coincident for a given x-axis time. The "SAMPLING" algorithm finds which curve has the least number of points, and then chooses the same number of points from the second curve, choosing the points closest to the desired "base" curve. This sampling can create "aliasing" problems in rapidly varying data, therefore the user should refer to the boxes drawn when the analysis is performed on less than 50 data points. In addition to the linear regression coefficients, the arithmetic mean (average) and the standard deviation are displayed alongside each curve.

Help Option

Pressing the "H" key invokes the help option. This option performs like its counterpart in the main screen mode, but details the commands available from the statistics mode.

Change Scaling

Pressing the "C" key invokes the Change scaling option. The x or y axis scaling of either statistics subplot (#1 or #2) may be changed, or the axes may be reset to the default scaling values. When changing scaling, the user is prompted for the axis and subplot information. The only significant departure between changing scaling in the statistics mode and changing scaling in the main PLOT mode is how the default scale values are presented. In the main plotting mode, the user is always presented a default value equal to the current axis limits the user is changing. In the statistics mode, however, the process is somewhat different. The main goal of changing scaling in the statistics mode is to allow the user to achieve identical x-axis durations on both subplots in order to obtain an auto- or cross-correlation. At the same time, however, the user might also want to examine all the data from one curve and compare it to a small section from another curve. To provide the flexibility needed, the user is always presented default XMAX values on the x-axis that result in identical durations between the subplot the user is changing and the other subplot. If the default XMAX would result in exceeding the available data, however, the user is presented with a default equal to the end of the available data, and a warning is printed.

The y-axis scaling does not influence the statistics in any way. Changing y-axis scaling is provided only as a convenience to the user in order to help visualize the data being analyzed. It should be pointed out that correlation analysis is not affected by the scaling of the y-axis because any scale factor is incorporated in both the numerator and denominator of the correlation equation, thus cancelling its effect.

Update Display

As in the main display mode, the "U" key is used to signal the end of any scaling changes, at which time the screen may now update the curves, and if data is available for two identical x-axis durations, display the regression line for each curve and the correlation factor.

Display Numbers

The display numbers option of the statistics mode is different from its counterpart in the main plot mode in that it does not erase the entire screen to display a tabulation of 100 numbers. Instead, this option has been modified to leave the two regression subplots on the screen, prompt the user for which subplot to display numbers, and then display 40 (x,y) pairs using the lower half of the plotting screen. As with the normal mode display numbers option, the user may press the enter key to update to the next block of 40 numbers before the prior set has finished displaying. Pressing the "E" key after any given set of 40 values has been displayed allows the user to Exit the display numbers mode.

Exit

The "E" key is used to exit the statistics mode and return the user to the mode from which statistics was entered.

Next Plot File

The "P" key aborts the interaction with the current screen, and if another plotting screen has been defined, starts data collection for this next screen.

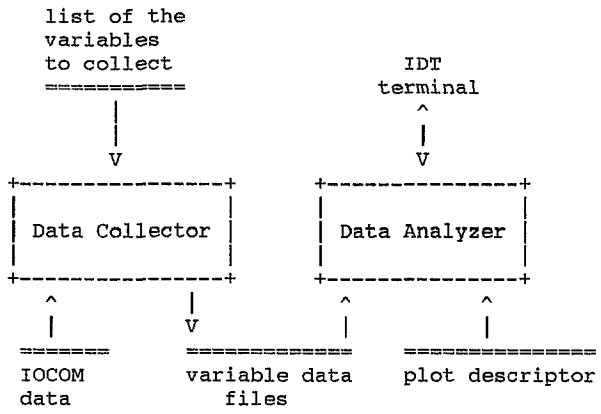
Abort to Menu

The "M" key aborts the current plot file interaction, and does NOT look for the next plot file, if any. Instead, this key aborts the entire function, returning the user to the menu (or DCL prompt) from which the program was entered.

3.3 Data Collection

The MPA Data Collector is an independent VAX function which collects data from the CSF (via IOCOM) and stores it on disk. The data collector reads an input definition file from the user to determine which variables are sampled and at what rate. The user may define up to 250 variables to be sampled once every second, and up to 250 more variables to be sampled once every minute. The data collector cycles every second, collecting IOCOM data into memory; the internal memory is then flushed out to the disk files after each sample is completed.

Due to the large amount of disk space consumed by this function, the user must be responsible for maintaining the disk files it creates. A maximum of seven days worth of data can be retained on the disk; at the start of the eighth day the disk will be full, and the data collector will fail. In order to retain the files indefinitely, the user must establish a procedure for moving the data files to tape before the disk fills up !

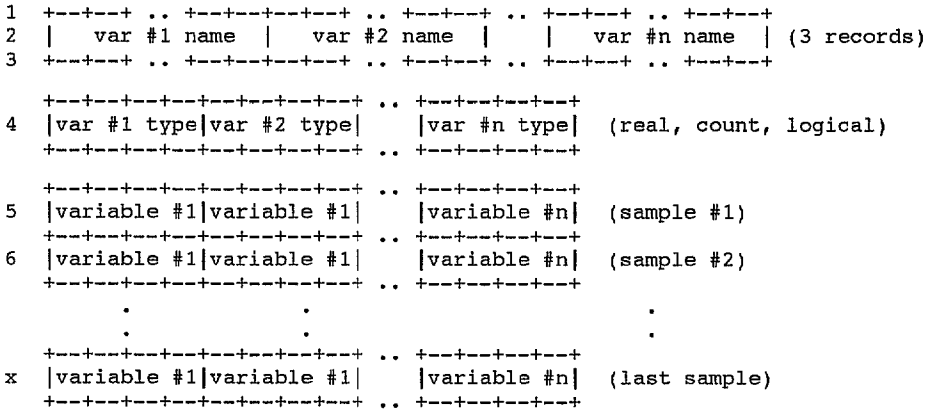


4 Maintenance

4.1 Data Collection Function

The data collection function maps to IOCOM and records the selected variables, independent of all other software. Because it maps to IOCOM through the Ada VMS_SECTIONS package, it is dependent on the IOCOM variable names only (not the actual layout of the common).

Mill Parameter Archiving File Layout



These files are direct access files. The records are (4 * the number of points being sampled) bytes long. Records 1-3 contain the ascii names of each CSF variable being sampled (12 bytes each). Record 4 contains the data types for each variable (an enumerated data type, four bytes each). The remaining records contain the samples from every variable for one sample period (each sample occupies 4 bytes, regardless of the type).

The data collector writes to two files; the fast_sample file and the slow_sample file. One record is written to the fast_sample file every second, and to the slow-sample file every minute. These files are located in the MPA_DATA$DIR directory and have file names which denote the date and time of the first sample in the file. For example, a fast_sample file might be called "10-APR-1988082330.FAST", meaning that it was opened at 8:23:30 on April 10, 1988.

Normally, there are 48 files (24 date.FAST and 24 date.SLOW) for each day, starting at midnight. New fast_sample and slow_sample files are opened every hour, to avoid creating one very large file (168,890 blocks). The exception is when the user changes the list of variables to be sampled and restarts the data collector; when this happens, the data collector opens new sample files from the current time. The maximum file size is 7,040 blocks for a complete hours worth of fast samples of 250 points, and 125 blocks for a complete hours worth of slow samples. Each day then consumes 171,960 blocks (slightly less then 1/7th of an RA82).

4.2 Data Analysis Function

The analysis function is an independent off line utility which does not interact with any other software on the system.

5 Configuration

The MPA Utility does not require any major configuration to install. This utility is provided with defaults that the customer may override. As discussed in the preceding sections, the selection of collection variables is customer driven, as is the layout and number of plot screens.

A Installation

The MPA function is distributed on tape as a save set containing all necessary object modules to build the system. The customer uses the DEC VAX/VMS command procedure "VMSINSTAL" to install the mill parameter archiving utility.